Barış Deniz Sağlam

Staff AI Engineer · agents, evals, architecture

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With 10+ years of engineering experience, I take AI products from first prototype to production and paying customers, and I have been the first engineer on new products several times. At Bosch I set the architecture of a multi-agent platform and built its core agent. As Novee's first engineer I built three LLM products, and all were sold. At Peppy I led the AI of an employee-benefits platform that was sold to enterprise clients and spun out as Euphoric. Earlier, I was the founding engineer of Supersense, an assistive app for blind users, and co-founded an ed-tech startup. Alongside engineering, I am doing a PhD on language models that learn from experience, and my open 27B document agent was a joint winner of the ICDAR 2026 DocVQA challenge.

EXPERIENCE

Staff AI Engineer · Independent Contractor

Mar 2025 – present · remote

Bosch Legends Lab · multi-agent platform for sourcing external experts

Jan 2026 – present

Architected a greenfield multi-agent platform around durable workflow execution and event-driven services, so agents run for hours at scale, survive failures and resume where they stopped, a design that held up where general-purpose agent harnesses stalled on long tasks.

Consolidated the platform's early microservices into a monolith until service boundaries were clear, which let a small team ship its first use cases quickly, then split it into services behind a gateway.

Built the core agent, a neuro-symbolic research agent that finds experts across the web and assesses them against a project's criteria, in runs of hours over up to ten thousand candidates. It outperforms other deep-research tools by a large margin on our benchmarks, and recruiters prefer it to their previous tools.

Set up the new team's engineering culture: RFCs for design decisions, CI/CD, API and observability standards, a roadmap of small milestone releases shaped with the PM and business manager, shared coding-agent workflows, and evaluation-driven agent development.

Novee AI · LLM products for insurance underwriting

May 2025 – Dec 2025

First engineer hired; worked directly with the CTO.

Built an LLM extraction product, delivered on a tight deadline, that won Novee a client contract: an ensemble of agents that pulls insured assets from messy PDFs, spreadsheets and email threads and reliably automates whole categories of submissions.

Built evaluation from day one and, with the PM and labeling team, fixed unreliable labels and product assumptions before optimising, so every design choice was decided by data.

Built a superhuman contract-comparison agent that caught material changes that underwriters missed. The labeling interface I built for its evaluation shipped to clients as the product's interface.

Designed self-correcting extraction agents that check their output against independent evidence in the documents and fix what fails, improving reliability substantially; other engineers reused the approach in other products.

Senior AI Software Engineer · Peppy Health

Mar 2023 – Feb 2025 · London (remote)

First engineer hired for AI.

Led the AI of an employee-benefits platform, from the prototype the business team tested with clients to a product sold to enterprise clients and spun out as Euphoric.

Shipped its benefits assistant agent, hybrid search and recommendations, choosing each algorithm by offline evaluation against expert labels.

Led most of the platform's backend architecture while the product was still taking shape, including a relational data model that absorbed many changes and an event-sourced Stripe checkout I built alone, safe to interrupt and resume, which other engineers extended to new product types.

Built personalised, event-triggered email campaigns that non-technical staff run themselves, from prototype to production.

Launched the health app's AI chatbot globally, building its backend and AI alone from idea to production, with clinician evaluation and safety guardrails for urgent and mental-health cases.

Mentored junior engineers through pair programming and gave the company's first knowledge-sharing session, on information retrieval.

Senior Software Engineer · Tractable Aug 2021 – Feb 2023 · London (remote)

Led cost-aware leakage detection end to end, the capability a prospective client required for Tractable's entry into a new European market; it turns model predictions into cost-optimal repair decisions to flag overcharging.

Built the human-in-the-loop review platform that routes uncertain model decisions to expert reviewers.

Brought ML models into production in a high-scale claims-processing system.

Pushed for CI/CD, which the organisation adopted, and mentored junior engineers and researchers.

Software Development Engineer · Amazon Web Services Oct 2020 – Jun 2021 · Berlin

Maintained and extended high-scale, low-latency pricing and offer services in AWS Billing; took part in on-call, wrote design docs and owned parts of projects.

Founding Engineer (ML and Mobile) · Mediate / Supersense Nov 2018 – Mar 2020 · Boston (remote)

As the first engineer at an MIT spin-off, built an on-device document-detection model end to end, from data collection and labeling to training and deployment; it guides blind users to frame documents with the camera. Also built an intent-recognition model on the camera feed and a synthetic-data pipeline for hard cases. Took the app, Supersense, from prototype to released Android and iOS apps.

Co-founder and Software Engineer · Comind Oct 2017 – Nov 2018 · Ankara

Co-founded and built a speech-based English-learning product for children: web app, serverless backend and user testing.

R&D Engineer, Structural Dynamics · Turkish Aerospace Industries Sep 2013 – Nov 2017 · Ankara

Built flight-test modal analysis software to validate aircraft structural models, replacing an expensive licensed tool. Wrote the department's onboarding handbook for engineers new to aerospace.

SELECTED WORK

ICDAR 2026 DocVQA, joint winner (8–35B category) · open 27B code agent that tells a vision model where to look · blog

Language of Learning, first author · position paper on models that learn from experience at test time · SSRN 2026

ArgGYM, third author · defeasible-reasoning benchmark, under review at ICLR 2027 · proposed turning the work into a benchmark, built and ran its evaluation · arXiv

Fine-tuning vs prompting for extraction · string-match metrics and LLM-judge ratings rank the two differently · dataset · blog

dspy-temporal, open source · durable execution for DSPy programs · GitHub

EDUCATION

PhD, Information Systems, Middle East Technical University (METU) · 2025 – present · research on language models that learn from experience

MSc, Data Informatics, METU · 2021 – 2024 · CGPA 3.93/4.00 · thesis on knowledge-graph-augmented multi-hop question answering with LLMs

BSc, Mechanical Engineering, METU · 2006 – 2012

SKILLS

AI	LLM agents and multi-agent systems, RAG and hybrid retrieval, evaluation design and LLM-as-judge, document extraction, RL post-training, PyTorch, Hugging Face
Backend	Python, TypeScript, Java; FastAPI, workflow orchestration (Temporal), event-driven systems (Kafka, NATS), event sourcing, PostgreSQL, Redis
Infrastructure	AWS, Azure, GCP, Kubernetes, Docker, CI/CD, OpenTelemetry
Languages	Turkish (native), English (professional)